# DOKUMENTACIJA: ASP.NET CORE WEB API

## 1. Uvod

**ASP.NET Core Web API** je backend tehnologija iz Microsoft .NET ekosistema koja služi za pravljenje REST API-ja.
Pomoću nje možeš da praviš aplikacije koje primaju HTTP zahteve (GET, POST, PUT, DELETE) i vraćaju podatke najčešće u **JSON** formatu. ASP.NET Core podržava izradu API-ja pomoću **Controllers** pristupa, ali i pomoću **Minimal API** pristupa.
Primeri upotrebe:

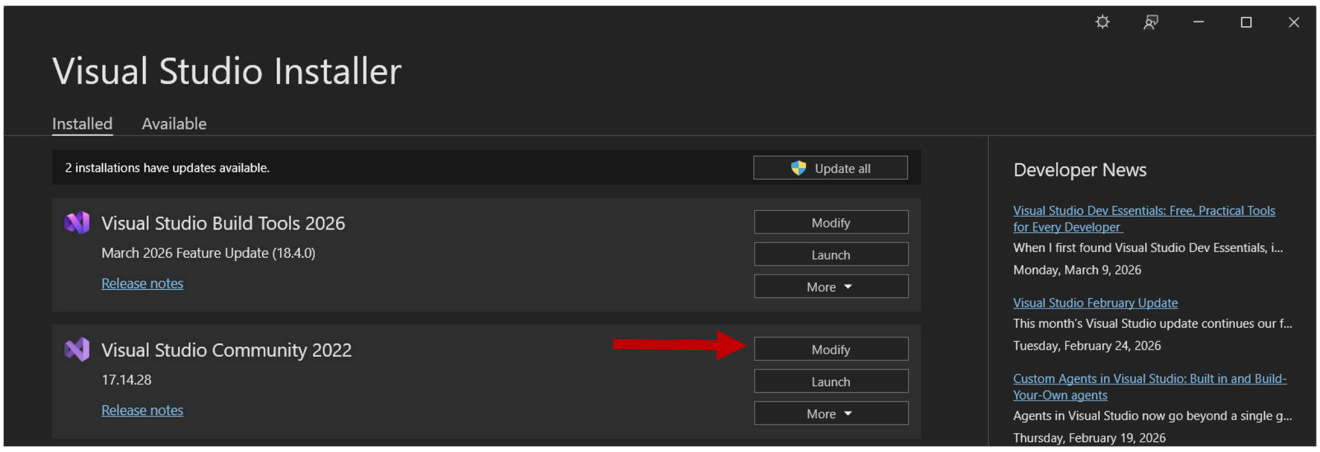
- backend za **Angular**, React ili Vue aplikaciju
- mobilni backend
- sistem za autentikaciju i korisnike
- CRUD aplikacije nad bazom

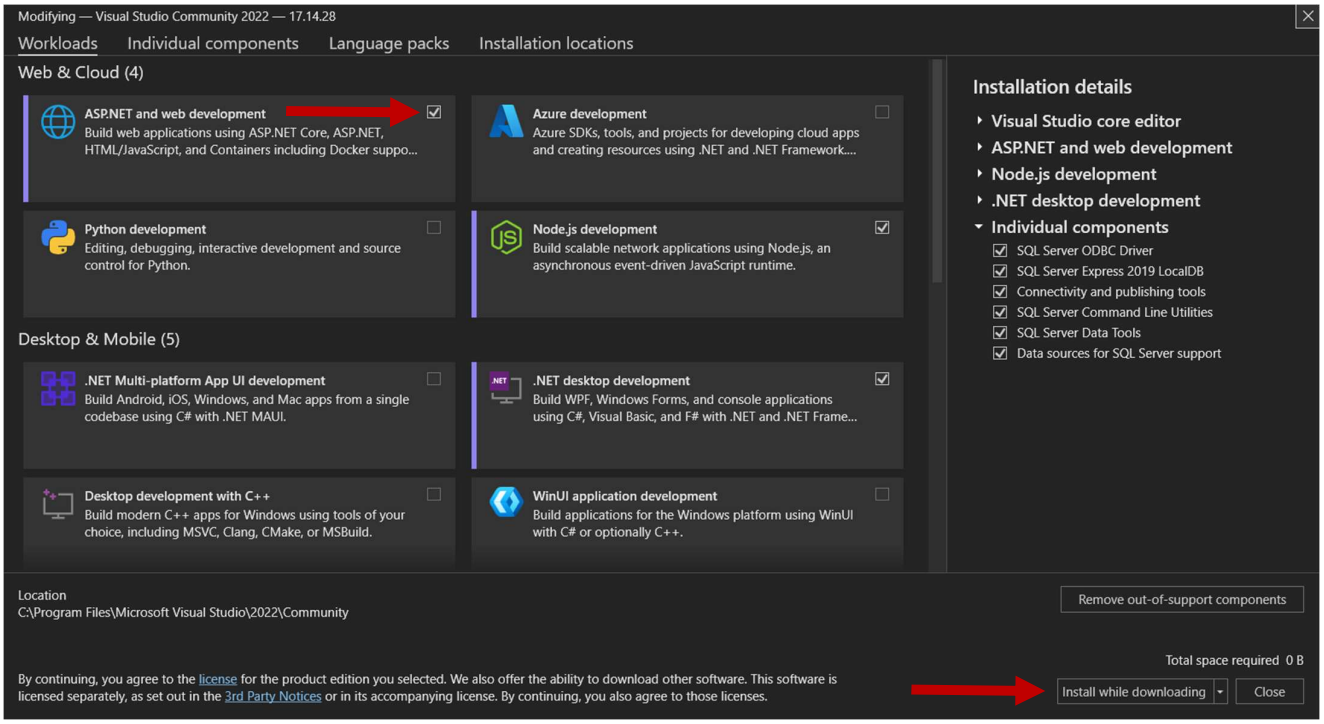
## 2. Šta je potrebno za rad

Da bi mogao da razvijaš ASP.NET Core Web API projekat, potrebno je da imaš:

- **.NET SDK** instaliran na računaru (trenutna verzija .NET 10.0), jer SDK omogućava kreiranje, build i pokretanje .NET projekata. Code editor ili IDE, najčešće:
    - **Visual Studio 2026** (besplatna Community verzija)
    - **Visual Studio Code**
- po potrebi:
    - **Postman/Swagger** alat za testiranje API-ja
    - baza podataka, **PostgreSQL…**

Ako radiš u **Visual Studio 2026**, potrebno je da tokom instalacije bude uključen workload **ASP.NET and web development**. Microsoft to navodi kao obavezan preduslov za ASP.NET Core razvoj u Visual Studio okruženju.






## 3. Kreiranje projekta u Visual Studio-u

**Korak 1:** Otvori **Visual Studio 2026** i klikni na **Create a new project**.

**Korak 2:** U pretrazi šablona ukucaj: **ASP.NET Core Web API (**izaberi taj template i klikni **Next**.)

**Korak 3:** Unesi:

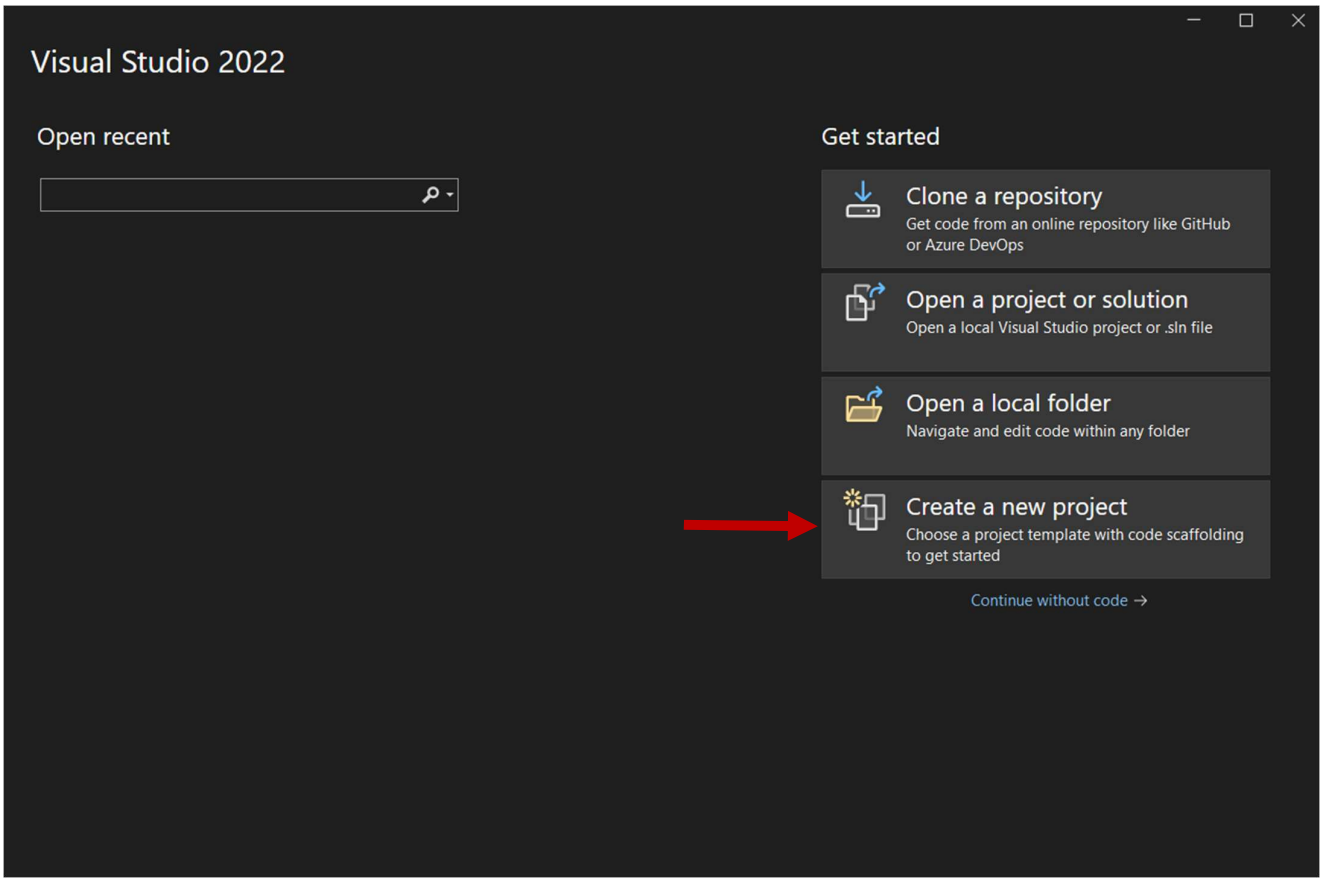
- **Project name**
- **Location**
- **Solution name**

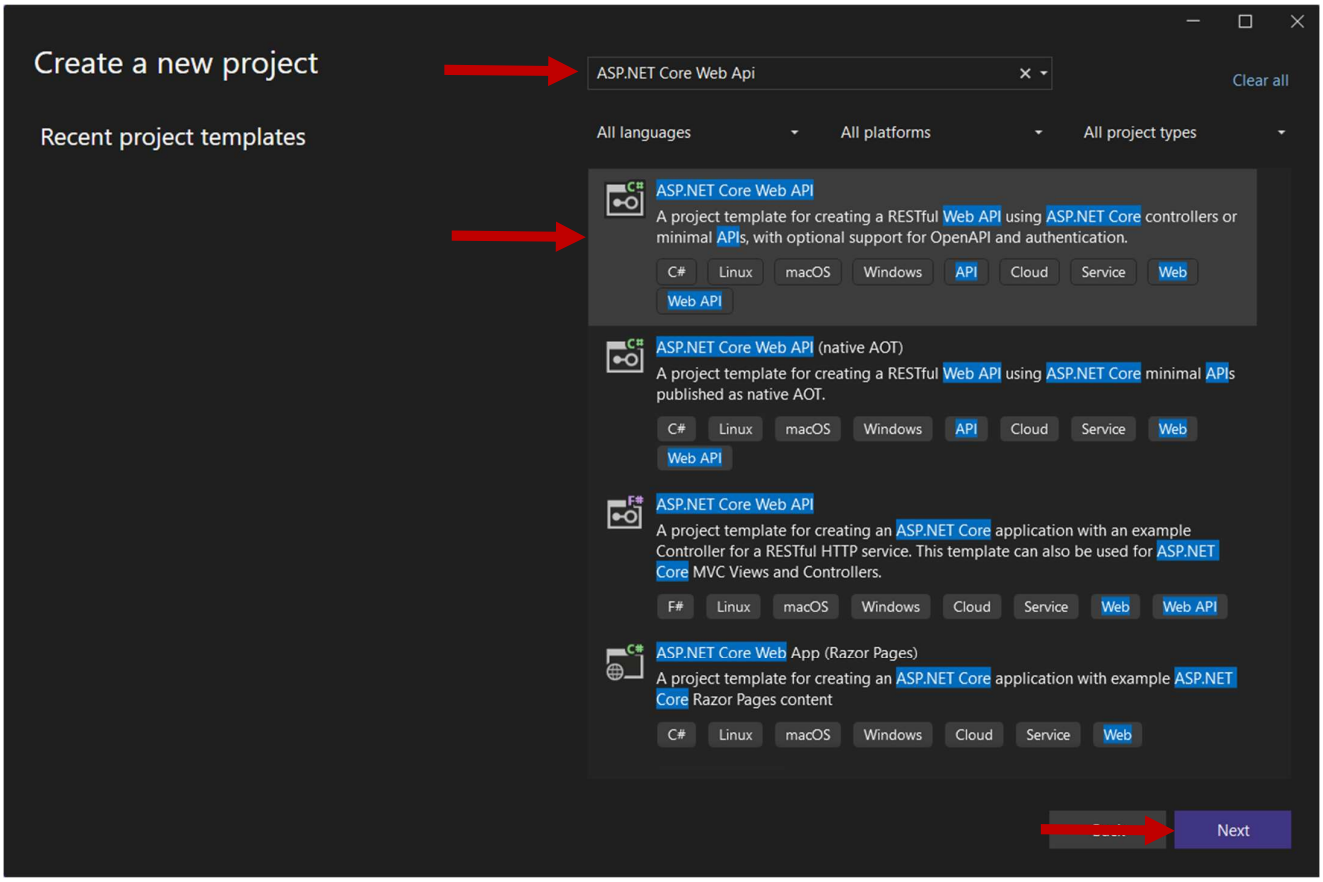
Klikni **Next**.

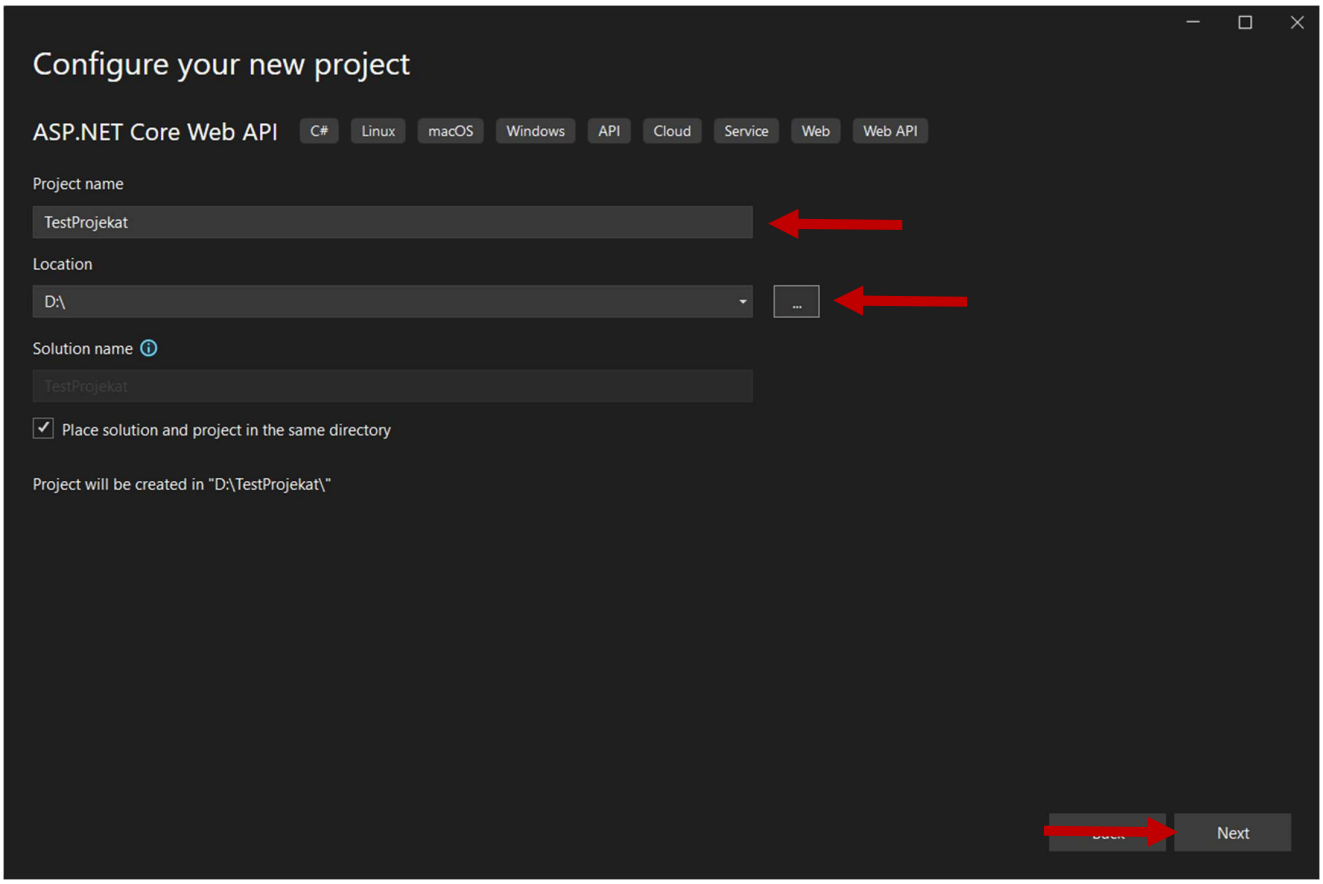
**Korak 4**

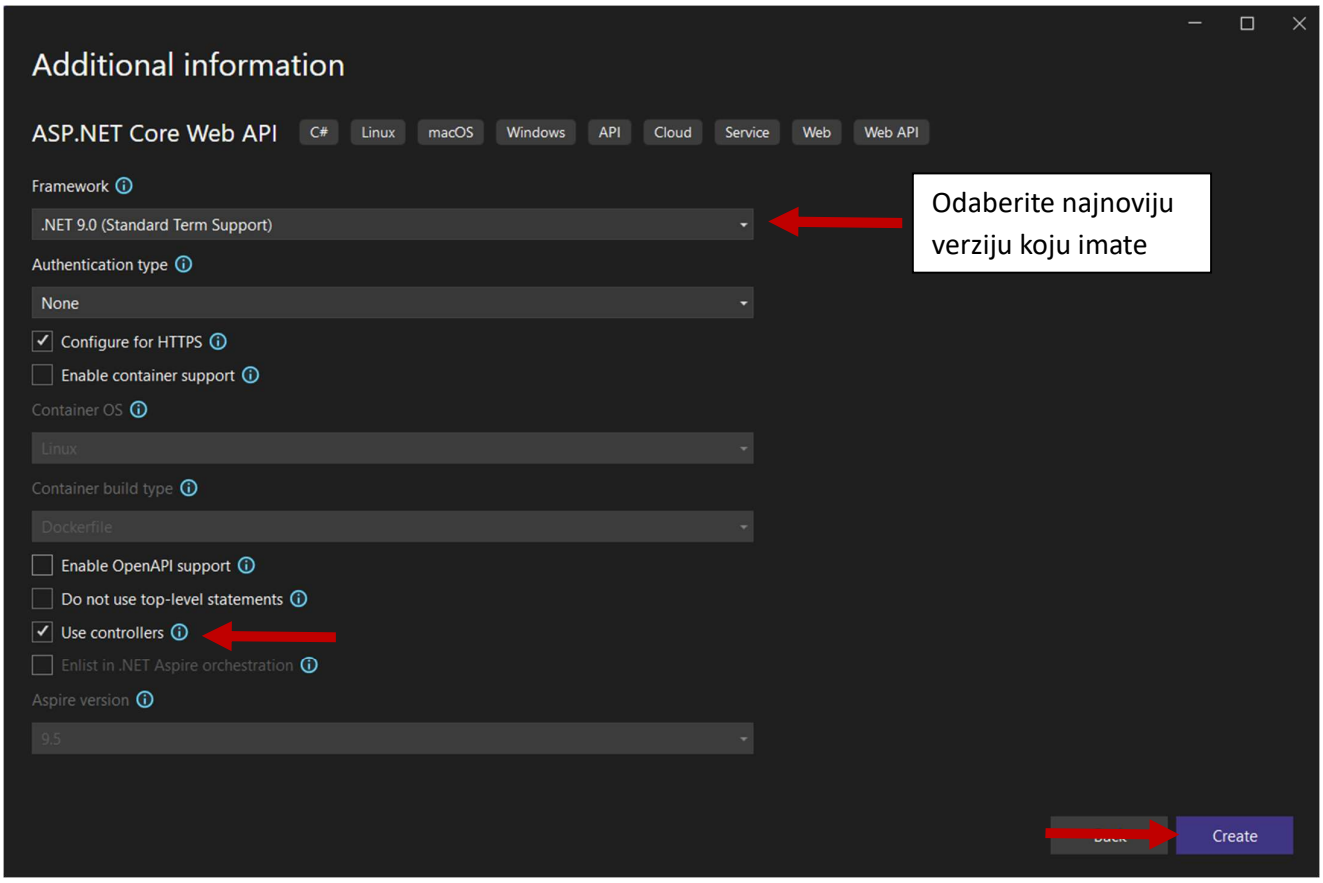
U dodatnim podešavanjima biraš:

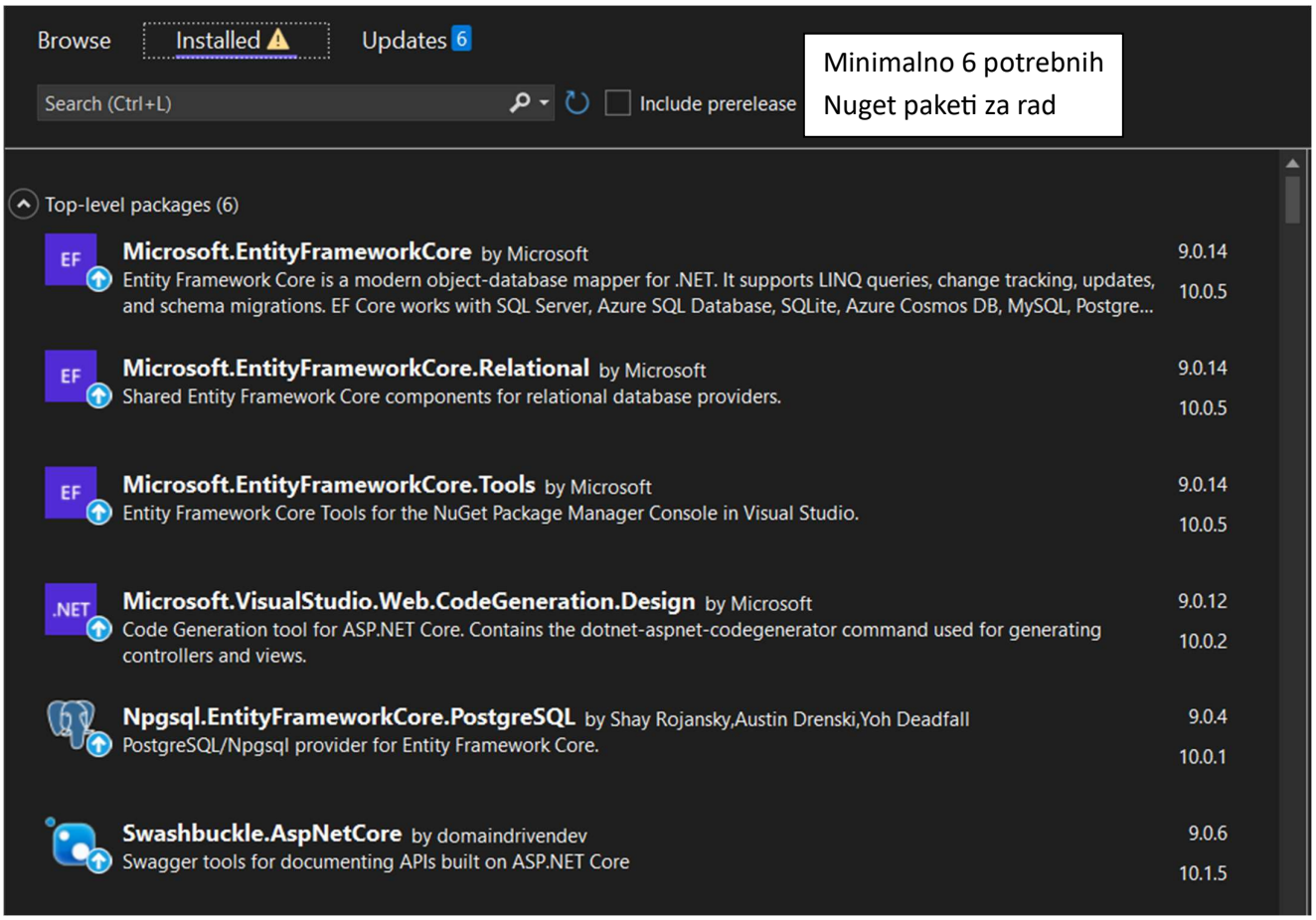
- .NET verziju
- da li želiš OpenAPI/Swagger podršku
- da li želiš Controllers pristup
- autentikaciju ako je potrebna


Visual Studio 2022
Open recent
Get started
Clone a repository
Get code from an online repository like GitHub or Azure DevOps
Open a project or solution
Open a local Visual Studio project or .sln file
Open a local folder
Navigate and edit code within any folder
Create a new project
Choose a project template with code scaffolding to get started
Continue without code →


Create a new project
ASP.NET Core Web Api
Clear all
Recent project templates
All languages
All platforms
All project types
ASP.NET Core Web API
A project template for creating a RESTful Web API using ASP.NET Core controllers or minimal APIs, with optional support for OpenAPI and authentication.
C# Linux macOS Windows API Cloud Service Web Web API
ASP.NET Core Web API (native AOT)
A project template for creating a RESTful Web API using ASP.NET Core minimal APIs published as native AOT.
C# Linux macOS Windows API Cloud Service Web Web API
ASP.NET Core Web API
A project template for creating an ASP.NET Core application with an example Controller for a RESTful HTTP service. This template can also be used for ASP.NET Core MVC Views and Controllers.
F# Linux macOS Windows Cloud Service Web Web API
ASP.NET Core Web App (Razor Pages)
A project template for creating an ASP.NET Core application with example ASP.NET Core Razor Pages content
C# Linux macOS Windows Cloud Service Web
Next


Configure your new project
ASP.NET Core Web API
C#
Linux
macOS
Windows
API
Cloud
Service
Web
Web API
Project name
TestProjekat
Location
D:\
...
Solution name
TestProjekat
Place solution and project in the same directory
Project will be created in "D:\TestProjekat\"
Next


Additional information
ASP.NET Core Web API
C#
Linux
macOS
Windows
API
Cloud
Service
Web
Web API
Framework
.NET 9.0 (Standard Term Support)
Odaberite najnoviju verziju koju imate
Authentication type
None
Configure for HTTPS
Enable container support
Container OS
Linux
Container build type
Dockerfile
Enable OpenAPI support
Do not use top-level statements
Use controllers
Enlist in .NET Aspire orchestration
Aspire version
9.5
Create


Browse
Installed
Updates 6
Search (Ctrl+L)
Include prerelease
Minimalno 6 potrebnih
Nuget paketi za rad
Top-level packages (6)
Microsoft.EntityFrameworkCore by Microsoft
Entity Framework Core is a modern object-database mapper for .NET. It supports LINQ queries, change tracking, updates, and schema migrations. EF Core works with SQL Server, Azure SQL Database, SQLite, Azure Cosmos DB, MySQL, Postgre...
9.0.14
10.0.5
Microsoft.EntityFrameworkCore.Relational by Microsoft
Shared Entity Framework Core components for relational database providers.
9.0.14
10.0.5
Microsoft.EntityFrameworkCore.Tools by Microsoft
Entity Framework Core Tools for the NuGet Package Manager Console in Visual Studio.
9.0.14
10.0.5
Microsoft.VisualStudio.Web.CodeGeneration.Design by Microsoft
Code Generation tool for ASP.NET Core. Contains the dotnet-aspnet-codegenerator command used for generating controllers and views.
9.0.12
10.0.2
Npgsql.EntityFrameworkCore.PostgreSQL by Shay Rojansky,Austin Drenski,Yoh Deadfall
PostgreSQL/Npgsql provider for Entity Framework Core.
9.0.4
10.0.1
Swashbuckle.AspNetCore by domaindrivendev
Swagger tools for documenting APIs built on ASP.NET Core
9.0.6
10.1.5

## 4. Osnovna struktura projekta

Tipičan ASP.NET Core Web API projekat sadrži:

```
[NazivProjekta]/
│
├── Controllers/
├── Properties/
├── appsettings.json
├── appsettings.Development.json
├── Program.cs
├── MyApi.csproj
└── WeatherForecast.cs
```

**Objašnjenje:**

**Controllers/:** Ovde se nalaze klase koje obrađuju HTTP zahteve.

**Program.cs:** Glavna ulazna tačka aplikacije.
Tu se konfigurišu:

- servisi
- dependency injection
- middleware
- rutiranje
- Swagger
- CORS
- konekcija ka bazi

**appsettings.json:** Konfiguracioni fajl za:

- connection string
- logovanje
- custom podešavanja aplikacije

**.csproj:** Projektni fajl koji definiše:

- target framework
- Nuget pakete
- build podešavanja

## 5. Šta dobijaš u startnom projektu

Kada kreiraš novi Web API projekat, obično dobiješ:

- primer endpoint-a
- osnovni Program.cs
- Swagger/OpenAPI konfiguraciju
- development HTTPS setup

ASP.NET Core šabloni standardno uključuju podršku za OpenAPI/Swagger kada je ta opcija uključena, što olakšava testiranje API-ja kroz browser interfejs.

**U novijim verzijama ASP.NET Core aplikacije, većina konfiguracije je smeštena u Program.cs.**

**Objašnjenje:**

- [ApiController] označava da je klasa API kontroler
- [Route("api/[controller]")] formira rutu, npr. api/users
- [HttpGet] odgovara na GET zahteve
- Ok(users) vraća HTTP 200 i JSON podatke

## 6. Pokretanje i testiranje API-ja

**Pokretanje**

U terminalu: **dotnet run** ili u Visual Studio-u: klikom na **Run** / **F5**.

**Testiranje – Swagger:** Kada pokreneš projekat, često se otvori Swagger stranica:

https://localhost:xxxx/swagger

Tu možeš:

- videti sve endpoint-e
- slati zahteve
- testirati odgovor servera

## 7. HTTP metode u Web API-ju

Najčešće metode:

**GET:** Koristi se za čitanje podataka.

**GET by id:** Koristi se za čitanje podataka po **id**.

**POST:** Koristi se za dodavanje novog resursa.

**PUT:** Koristi se za izmenu postojećeg resursa.

**DELETE:** Koristi se za brisanje resursa.

## 8. Modeli

Model predstavlja podatke sa kojima radi aplikacija.

Primer:

```
namespace [DemoProjekat].Models
{
   public class User
   {
      public int Id { get; set; }
      public string Name { get; set; } = "";
      public string Email { get; set; } = "";
   }
}
```

## 9. appsettings.json i connection string

Primer:

```
{
  "ConnectionStrings": {
    "DefaultConnection":
"Host=localhost;Port=5432;Database=NazivKataloga;Username=postgres;Password=root"
  }
}
```

**U Program.cs možeš povezati bazu i DbContext.**